

Week 1: Generative AI & LLM Foundations

Learning Topics

- Introduction to Artificial Intelligence (AI)
- Introduction to Generative AI
- Evolution of Large Language Models (LLMs)
- Transformer Architecture (Overview)
- Tokens & Tokenization
- Context Window
- Temperature, Top-p & Max Tokens
- Prompt Engineering Fundamentals
- Zero-shot, One-shot & Few-shot Prompting
- OpenAI API Basics
- Gemini API Basics
- Claude API Basics
- API Keys & Environment Variables
- Python Virtual Environment
- Git & GitHub Basics

Weekly Task

Build an **AI Chat Assistant** using any one of the following LLM APIs (**OpenAI, Gemini, or Claude or any Free of cost LLMs**) that can answer user queries through either a **simple Command-Line Interface (CLI)** or a **Web Interface**.

Expected Deliverables

- Complete source code
- GitHub repository
- README.md with setup instructions
- requirements.txt
- Environment variables configured using a .env file
- Working AI Chat Assistant capable of responding to user queries

Submission

Submit the **GitHub repository link** before the assigned deadline. Ensure the repository includes all required files and documentation.